

b) write stuff in $K_A \rightarrow \beta \rightarrow (\overrightarrow{BA}, \overrightarrow{BB}, \overrightarrow{BC})$ using $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$.

Lecture 1 - Graphs

24.02.2026

- max 1 absence for labs
- max 2 abs. for seminars
- An cheat sheet $\rightarrow$ to be turned in after exam
- ≥ 4.5 for labs, ≥ 4.5 for written exam

Why do we need graphs? - 4 kinds of graphs $\leftarrow$ directed

vertex - vertices

X = set of vertices (váršuvu)

- non-empty
- finite

E = set of edges (muchi)

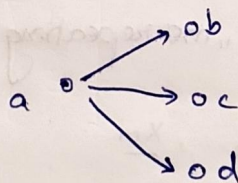
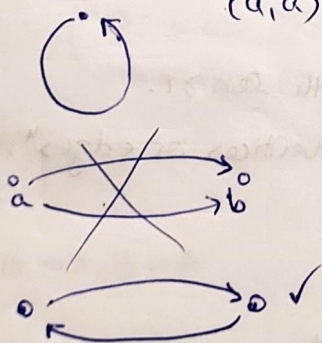
Directed graph:

$$E \subseteq X \times X$$

$(a, b) \in E$ ordered pair.

package "a" depends on package "b"

(a, a) loops $\rightarrow$ not formally forbidden, but E is a set, so we can't have equal edges.



Undirected graphs:

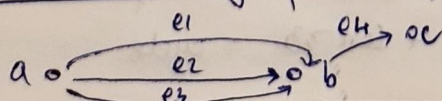
$$E \subseteq \{ \{x, y\} \mid x, y \in X, x \neq y \}$$

$\rightarrow$ loops are forbidden.

$\rightarrow$ harder to work with

$\rightarrow$ good for trees.

Directed multi-graph : $X, E, f: E \rightarrow X \times X$



set of pairs of vertices

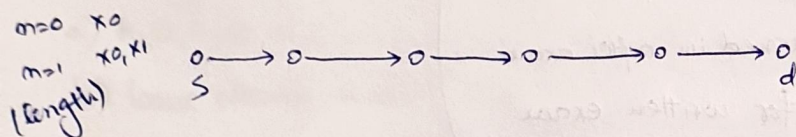
$$f(e_1) = (a, b)$$

$$f(e_2) = (a, b)$$

$$\dots$$

$$f(e_n) = (b, c)$$

walk $\rightarrow$ context of graph $\rightarrow$ a ^{finite} sequence $(x_0, x_1, \dots, x_m)$; $x_0, \dots, x_m \in X$
 $(x_i, x_{i+1}) \in E$
 $(\forall) i \in \{0, \dots, m-1\}$
 $\{x_i, x_{i+1}\} \in E.$



path $\rightarrow$ walk with no repeating vertices or edges
 (if vertices don't repeat, edges don't repeat) $\rightarrow$ redundancy

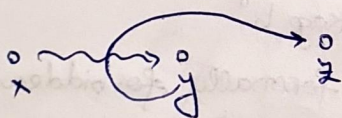
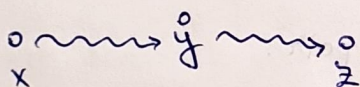
Graph = Binary Relation (algebra) = the set of edges $\rightarrow$ is a relation

Transitive closure of the first graph (accessibility relation):

$$(x, y) \in E'$$

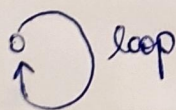
is transitive because we use the term "walk", not "path".

$$\exists \text{ walk} \Rightarrow x = x_0, x_1, \dots, x_m = y$$



Closed walk = walk with start = end with $\text{len} \geq 1$.

cycle = closed walk with "no repeating vertices or edges".

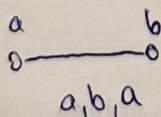


$$\underline{x_0, x_1, \dots, x_m}$$

$$x_0 = x_m$$

$$x_0, \dots, x_{m-1} \text{ distinct}$$

x_m
 (x_{m-1}, x_m)
 (x_{m-1}, x_m) $\rightarrow$ this edge may repeat itself.



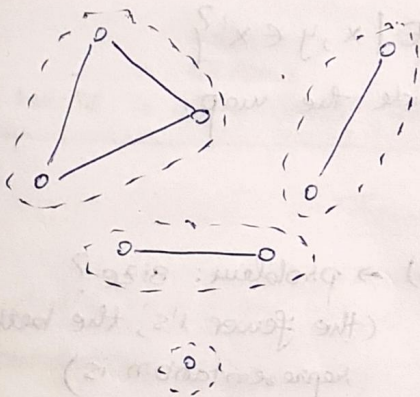
Tree = connected graph w no cycles.
 undirected case + $\text{len} \geq 3$ = cycle.

Connected component $\rightarrow$ undirected graph

connected graph = each vertex is accessible from each vertex.

strongly connected graph = $\rightarrow$ directed graph

$\circ \leftarrow \circ \rightarrow \circ$ (not strongly connected)



Connected component =

(componentă conexă) $\rightarrow$ take a vertex and see the ~~subset~~ vertices that are linked to it $\Rightarrow$ subset $\Rightarrow$ comm. comp.

Equiv. rel $\rightarrow$ reflexive ($A=A$)

$\rightarrow$ transitive ($A=B, B=C \Rightarrow A=C$)

$\rightarrow$ symm. ($A=B \Rightarrow B=A$)

$\rightarrow$ homog. rel. ($e: A \rightarrow A$)

$$\begin{array}{l} A=B \\ B=C \end{array} \mid \Rightarrow A=C$$

$$A=B \Rightarrow B=A$$

Objects = vertices

In order to define components, you need equiv. relation.

Directed graph $\rightarrow$ strongly connected components

$a \sim b$ if $\left\{ \begin{array}{l} b \text{ is accessible from } a \\ a \text{ is accessible from } b \end{array} \right.$
relation

Subgraphs $\rightarrow$ map of the city is a subgraph of the map of the country.

$G=(X,E)$ $\rightarrow$ remove vertices along with their edges!

$$G'=(X',E') \quad X' \subseteq X \\ E' \subseteq E$$

• Spanning subgraph: $X=X'$

• Induced subgraph: $E' = \{(x,y) \in E \mid x,y \in X'\}$

$\hookrightarrow$ Keep the streets; remove those outside the map.

Internal representation of a graph

• adjacency matrix (matrice de adiacență) $\rightarrow$ problem: size?

$i \rightarrow$

	j
	1
	0

1: $(i,j) \in E$

0: $(i,j) \notin E$

(the fewer 1's, the better this representation is)

! a graph with n vertices: $|X|=n$

$$|E| \leq n^2$$

How Zip works: $E \rightarrow$ more frequent $\rightarrow$ shorter code
 $B \rightarrow$ less frequent $\rightarrow$ longer code $\} \Rightarrow$ optimization

Sparse vs Dense graphs.